

20250310EB_MN (CellTiter-Glo, RNA, imaging)

Project: Yi-Shuan Tseng Holt Lab

Author: Yi Shuan Tseng

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2025年3月10日 星期一

Use Neurobasal-A without glucose and sodium pyruvate for experiment. I didn't add extra Na pyruvate this time.

Cells with low or no glucose died. All cells look unhealthy under the microscope. Do not use this batch

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年3月12日 星期三

T2 -> T3

2025年3月14日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年3月17日 星期一

T4 -> T5

2025年3月19日 星期三

T5 -> T6

2025年3月21日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年3月24日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.

4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

Table3

	A	B	C	D	E	F	G	H
1	conc. (M/mL)	survival	total (5mL)	well-type	well number	K/well	volume for one well (uL)	
2	1.4	99	7	6-well	3	1200.0	857	
3				24-well	12	233.3	500	
4				96-well	12	50.0	36	

Table4

	A	B
1	condition #	
2	1	2.5mM 1h
3	2	0.15mM 1h
4	3	0.15mM 24h
5	4	2.5mM 24h

plate A

	1	2	3
A	1	2	3
B			

plate B (blue cells are from Chao)						
	1	2	3	4	5	6
A	no virus	vLH110 (12uL in total)	vLH111 (12uL in total)	vLH92 5uL	vIH129 (2uL/well)	
B				10uL		
C				20uL		
D				50uL		



plate C												
	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												



2025年3月25日 星期二

- T7 -> T7 for plate B
- B4, C4, D4 on plate B died (with black particles in media -> contamination?)

2025年3月26日 星期三

T7-> T7 for plate A, C

2025年3月27日 星期四

T7 -> BP T8

2025年3月29日 星期六

- 1/2 BP T8 -> 1/2 BP T8
- for plate C, just add 100uL fresh media directly

2025年3月31日 星期一

1/2 BP T8 -> 1/2 BP T8

2025年4月2日 星期三

1/2 BP T8 -> 1/2 BP T8
prepare media with Neurobasal-A w/o glucose

Table1					
	A	B	C	D	E
1		glu stock (M)	final concentraion (mM)	final volume	added glu volume (uL)
2	2.5mM	1.1	2.5	40	90.9
3	0.15mM	1.1	0.15	40	5.5

Treat plate C: C1-3, D1-3 at 4:00pm
Treat plate A: A3 at 7:30pm

2025年4月3日 星期四

Celltiter-Glo assay (plate C)

try to take BF images before the experiment for cell count, but cytation could not get focus -> use EVOS instead
24h treatment of both 2.5mM and 0.15mM were mostly died except for C1. I think it might related to the force when I added the media since I use the seronogical pippete. Also, the cells might not be really healthy according to the condition of the plate B on Friday. Do not use the data from C1-3 and D1-3.
Treat plate C: A1-3, B1-3 at 3pm.

prepare ATP calibration					
	A	B	C	D	E
1	A	B	C	D	E
2	100mM	1mM	1uM	100nM	10nM
3		100	1000	10	10
4		1uL A in 99uL media	1uL B in 999uL	50uL C in 450uL media	50uL D in 450uL media

plate C_glo assay													^
	1	2	3	4	5	6	7	8	9	10	11	12	
A	1			C in 2.5mM media									
B	2			D in 2.5mM media									
C	3			E in 2.5mM media									
D	4			C in 0.15mM media									
E	2.5mM media only			D in 0.15mM media									
F	0.15mM media only			E in 0.15mM media									
G													
H													

readout													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1		1	2	3	4	5	6		condition #		ave	-media background	2.5mM ATP conc.
2	A	13732	15366	12042	67723	66202	66291		1	2.5mM 1h	13713.333	13587.000	197.0058653099
3	B	6109	6129	2726	7156	6904	7957		2	0.15mM 1h	4988.000	4902.667	66.989524918
4	C	3273	461	454	1234	1486	1285		3	0.15mM 24h	died		
5	D	289	278	566	77597	77204	71873		4	2.5mM 24h	died		
6	E	73	101	205	7237	7369	6945			2.5mM media only	126.333		
7	F	79	71	106	915	947	994			0.15mM media only	85.333		
8	G								in 2.5mM	1uM	66738.666666667	equation: y=66.038x+703.46	
9	H									100nM	7339		
10										10nM	1335		
11									in 0.15mM	1uM	75558	equation: y=75.608x-76.944	
12										100nM	7183.666666667		
13										10nM	952		

RNA extraction (plate A)

Treat plate A: A1,2 at 6:30pm.

wash plate with cold PBS and add 1mL Trizol for each well -> freeze samples in -80

Imaging prep (plate B)

Treat plate B: according to the time table below (24h time point)

Table2								
	A	B	C	D	E	F	G	H
1			110 (column 2)		111 (column3)		TDP43 (column 1)	
2	condition #		treat	image	treat	image	treat	image
3	1	2.5mM 1h	9:18 (10:30 1/3000SirDNA)	11:20	12:10	13:00	13:30	14:40
4	2	0.15mM 1h	9:40 (10:30 1/3000SirDNA)	11:00	12:45	14:30	13:30	14:30
5	4	2.5mM 24h	11:30 4/3 (10:30 1/3000SirDNA)	11:55	14:30 4/3 (12:10, 12:45 with 1/3000SirDNA)	14:10	16:30 4/3 (15:00 with JFX and SPY595)	16:00
6	3	0.15mM 24h		11:30		14:40		16:30 (died)

plate B condition						
	1	2	3	4	5	6
A	1	1	1			
B	2	2	2			
C	4	4	4			
D	3	3	3			

2025年4月4日 星期五

Imaging plate B

Cells were unhealthy.

import reporters are not moving after activation in the most cells. Only see one or two cells wth moving reporter

export reporters moved, but they could not reset. The import.system in cells might be ruined.

The DNA staining are mostly perinucelaer staining and nuclei were round.

cells with 24h hypoglycemia died

cells also died after imaging

-> did not finish reporter eperiments.

-> I still imaged A1, B1, and C1; however, cells also looked unhealthy and may consider not using the images